

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3.	(a)			- Mechanical digestion {breaks food down into smaller pieces/ increases its (total) surface area} (1) Reject molecules - Chemical digestion involves {hydrolysis/ use of enzymes/ larger into smaller <u>molecules</u>/ insoluble into soluble <u>molecules</u>} (1)	2			2		
	(b)	(i)		Both correct for one mark Lipase = C Accept pancreas Bile = H Accept liver	1			1		
		(ii)		Help to neutralise the acid(ic) {chyme/ from the stomach} (1) Help to provide the {optimum pH/ alkaline (environment)} for the enzymes (in the duodenum) / prevents denaturing of enzymes (1)	2			2		
	(c)	(i)		To ensure that the {pH was {above/at} pH 10 / the contents of the tubes were pink} (at the start of the experiment) (1)		1		1		1
		(ii)		To ensure that the <u>concentration</u> of {lipids/ sodium carbonate solution/ phenolphthalein} remained the same (in each test tube) (1)		1		1		1

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	(d)	(i)		Any four (×1) from: A. (In tube A and B) the lipase {hydrolyses/ breaks down} the lipids (1) B. fatty acids {lower the pH/ make the solution more acidic} (1) C. (lowering the pH causes) (the phenolphthalein to) decolourise/ or description of (1) D. In tube B the bile salts {emulsifies / or description of} (1) E. Which creates a larger (total) surface area of the lipids (1) F. {Hydrolysis/ fatty acid production} occurs at a faster rate (1) ignore decolourises faster		2	2	4		4
		(ii)		To show bile (salts) cannot {hydrolyse lipids/ cause the production of fatty acids} on its own/ to see if bile salts can hydrolyse lipids on its own / to show lipase is needed to hydrolyse lipids			1	1		1
		(iii)		Inaccuracy: results are {qualitative/ subjective/ or description of} / starting pH may have been different (1) Improvement: use a pH {meter/probe} /to record the actual change in pH over a fixed time period)/ use colorimeter / measure {light transmission/absorbance by the solution} (1)			2	2		2
				Question 3 total	5	4	5	14		9